Sudam Kumar Paul

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LinkedIn Profile
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Experience

- **AI/ML Engineer** *Jan, 2026–Present*
  Multi-agentic Conversational Assistant for Retail Rezolve AI, Kolkata
  – Working on Agentic AI, MCP, and A2A (Agent-to-Agent) protocol based Voice Assistant.
  – Working on the next-generation, single-graph version of this assistant, where reason, plan and reflect agents share one scratchpad, with a per-tenant YAML tool registry, parallel guardrails, dynamic quick replies and real-time streaming to the frontend.
  – Developed a search-engine-agnostic REST service (FastAPI) that normalises text and image product search across multiple commerce search engines behind one contract, and worked on the knowledge-base search that answers store policy and FAQ questions.

- **IASc-INSA-NASI Summer Research Intern** *June-July 2025*
  Research Intern in Financial Economics Indian Institute of Technology, Kharagpur
  – Worked on Comparative Analysis of Asset Pricing Models: CAPM, APT and Fama-French Factors.

Projects

- **Multi-Agentic Conversational Retail Assistant** *Rezolve AI, Kolkata*
  A LangGraph-based multi-agent conversational AI shopping assistant: A2A and MCP version, and its single-graph successor Jan 2026–Present

  – Tools & technologies used: Python, FastAPI, LangGraph, Multi Agent, A2A Protocol, MCP, YAML, Langfuse, Weaviate, MongoDB, Redis, Docker.

  – Built a conversational AI shopping assistant where a model-agnostic, prompt-agnostic, and engine-agnostic LangGraph orchestrator routes natural-language queries to specialised downstream agents – recommendation, visual search, product interaction, FAQ, and cart operations – communicating over the A2A (Agent-to-Agent) JSON-RPC protocol with real-time SSE streaming to the frontend, while exposing live catalogue capabilities as typed MCP (Model Context Protocol) tool servers, with prompts managed via Langfuse for model-agnostic prompt versioning, per-user conversation memory via MongoDB checkpointing, and a centralised A2A Registry for agent discovery and health.
  – Working on the next-generation, single-graph version of this assistant, where reason, plan and reflect agents share one scratchpad instead of making agent-to-agent calls, with parallel safety guardrails and hard-capped tool loops. Tools, tone and search backend are configured per tenant in YAML, so onboarding a new client needs no code change. Also includes shopper memory across sessions, dynamic quick replies, a knowledge-base search for store policy and FAQ answers, and a search-engine-agnostic service that returns matching products to the frontend.

- **GauGan: Conditional Image Generation Using Gaussian-GAN** *Project Repository*
  GaussianGAN from scratch in TensorFlow/Keras using custom Generator, Discriminator and Encoder. March-May, 2025

  – Tools & technologies used: Python, TensorFlow, Keras, Google Colab, NumPy, Matplotlib, GANs, VAEs, SPADE (Spatially-Adaptive Normalization)

  – Implementing a conditional GAN model based on SPADE and VAE principles (GauGAN) to generate photorealistic images from semantic segmentation maps using the PASCAL_VOC_2012 and Facades datasets.

- **Implementation of ANN from Scratch** *Project Repository*
  Two-Layer Fully Connected Neural Network on MNIST dataset from scratch March-April 2025

  – Tools & technologies used: Python, Numpy, Matplotlib, Custom-built Neural Network, Cross-Entropy Loss & Softmax activation function, Gradient Descent & Momentum.

  – Designed and implemented a two-layer fully connected neural network from scratch using NumPy for handwritten digit classification on the MNIST dataset with sigmoid and softmax layers, trained via cross-entropy loss and gradient descent variants, achieving 96.92% train accuracy and 94.52% test accuracy with custom forward/backward propagations and manual parameter initialization.

- **Unsupervised Learning using Py-Spark** *Project Repository*
  Implementation of K_Means_Clustering in Distributed way using Py-Spark March-May 2025

  – Tools & technologies used: Java-21, Python 3.10, Pyspark 3.3.2, Numpy, Pandas, Matplotlib, VectorAssembler, Spark MLlib(KMeans), ClusteringEvaluator, Spark Cluster, Linux Environment, SparkSession.

– Developed a distributed K-Means clustering pipeline using PySpark on the Iris dataset, including feature engineering, scaling, and optimal cluster selection using Silhouette scores. Visualized cluster results with 3D scatter plots and evaluated model performance using PySpark's Clustering Evaluator.

- **Gas Turbine Energy Yield Prediction by Regression_Techniques** *Project Repository*
  Prediction of Turbine Energy Yield (TEY) using Various Regression_techniques August-September 2024
  – Tools & technologies used: Google Colab, Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn
  – A comparative study of various regression techniques, including linear regression, polynomial regression, gradient descent methods and regularization techniques to predict the Gas Turbine Energy and analyze their performance and computational efficiency on "Gas Turbine CO and NOx Emission Data Set_2015"
- **Comparative Analysis of Asset Pricing Models: CAPM, APT and Fama-French Factors** *Project Repository*
  Implementing and comparing traditional asset pricing models for portfolio performance analysis. June-July 2025
  – Tools & technologies used: Python (pandas, numpy, matplotlib, scikit-learn), Statsmodels: Linear Regression, Multivariate Regression
  – Performing a comparative analysis of CAPM, APT, and Fama-French models using regression techniques to evaluate their effectiveness in explaining asset returns.

Education

- **Ramkrishna Mission Vivekananda Educational and Research Institute** *2024-26*
  M.Sc. in Big Data Analytics (Data Science) CGPA: 7.64
- **Barasat Govt. College** *2020-23*
  B.Sc.(H) in Mathematics from West Bengal State University, Barasat, West Bengal Percentage: 87.45
- **Habra High School(H.S.)** *2020*
  West Bengal Council of Higher Secondary Education, West Bengal Percentage: 96.4

Technical Skills and Interests

- **Languages**: Python , R (Basic) , LaTeX
- **Agentic AI and Backend**: LangGraph , LangChain , FastAPI , REST APIs , MCP , A2A
- **Data Science and ML Tools**: NumPy , Pandas , Matplotlib , Scikit-learn , OpenCV
- **DL Frameworks**: Keras , TensorFlow , PyTorch
- **Tools**: Google Colab , Jupyter Notebook , VS Code , Git , Docker , Langfuse , Postman
- **Databases and Distributed Processing**: MongoDB , Redis , Weaviate , Neo4j , PySpark
- **Operating Systems**: Windows and Linux(Ubuntu)

Key Courses Taken

- **Data Structure and Algorithms using Python**
- **Artificial Intelligence and Machine Learning :** Machine Learning , Computer Vision , Deep Learning and it's application in NLP , Graph Machine Learning
- **Big Data Technologies :** Apache Spark , Graph Database- Neo4j , Data Mining
- **Mathematics for Computing :** Linear Algebra and Matrix Computation , Probability and Basic Statistics , Hypothesis Testing , Time Series and Survival Ananlysis , Multivariate Statistics , Introduction to Econometrics

Achievements

- **Successfully Completed a Course of Affective Computing in NPTEL** *Jan - April 2026*
  Affective Computing, offered by IIT Delhi
- **Successfully Completed a Course for Python Programming Language in NPTEL** *Aug - Sep 2025*
  Python for Data Science, offered by IIT Madras
- **Selected for Summer Intern at IASc-INSA-NASI** *June - July 2025*
  Summer Research Internship program 2025 in Financial Economics, IIT Kharagpur
- **Successfully Completed a Course of DSA using Python in NPTEL** *Jan - March 2025*
  Programming, Data Structures And Algorithms Using Python, offered by IIT Madras

Positions of Responsibility

- **Event Lead:** Perceptron 2026 Puzzle Event(Tech Fest at RKMVERI) *Jan, 2026*
- **Organizing Team Member:** Perceptron 2025 Puzzle Group (Tech Fest at RKMVERI) *Jan, 2025*

Hobby

- Reading Books, Watching Movies and Listening Audio Stories